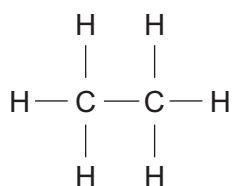
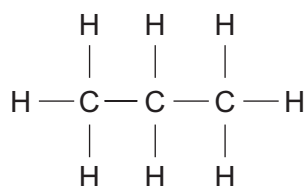
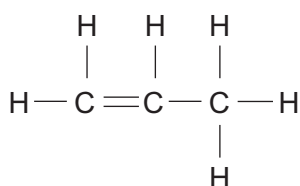
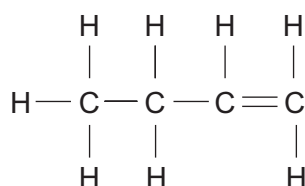


2. (a) The structural formulae of four carbon compounds are shown below.

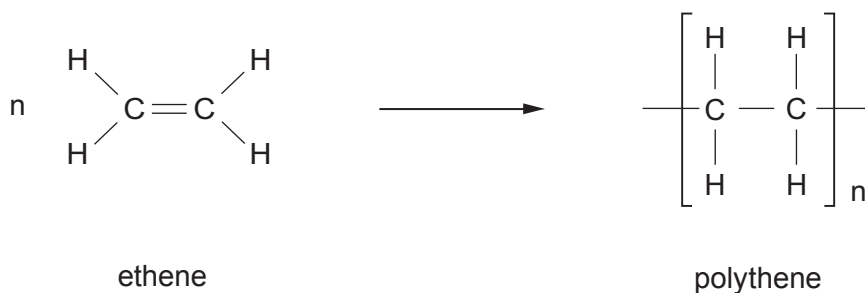
**A****B****C****D**

Complete the table by choosing the **letter A, B, C or D** which represents the structural formula of the named compounds. [2]

Name	Molecular formula	Structural formula
ethane	C_2H_6	
propene	C_3H_6	

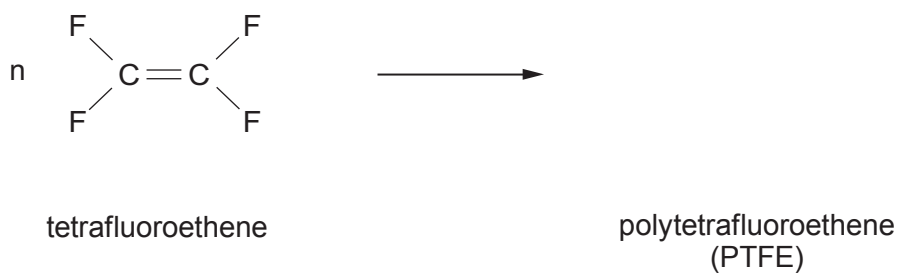


(b) The equation below shows the formation of polythene from ethene.

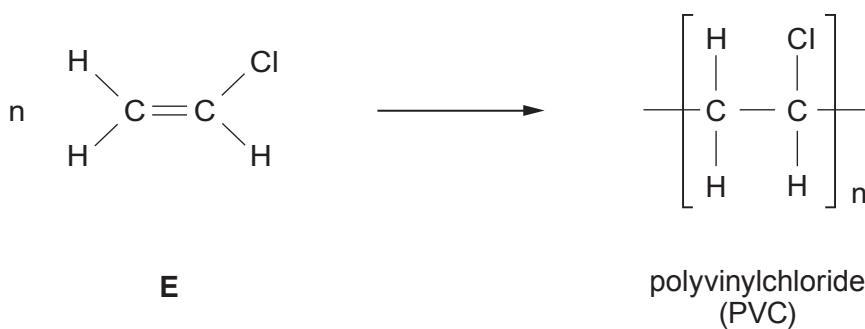


Use this information to complete parts (i) and (ii).

(i) Complete the equation for the formation of polytetrafluoroethene (PTFE) from tetrafluoroethene. [1]



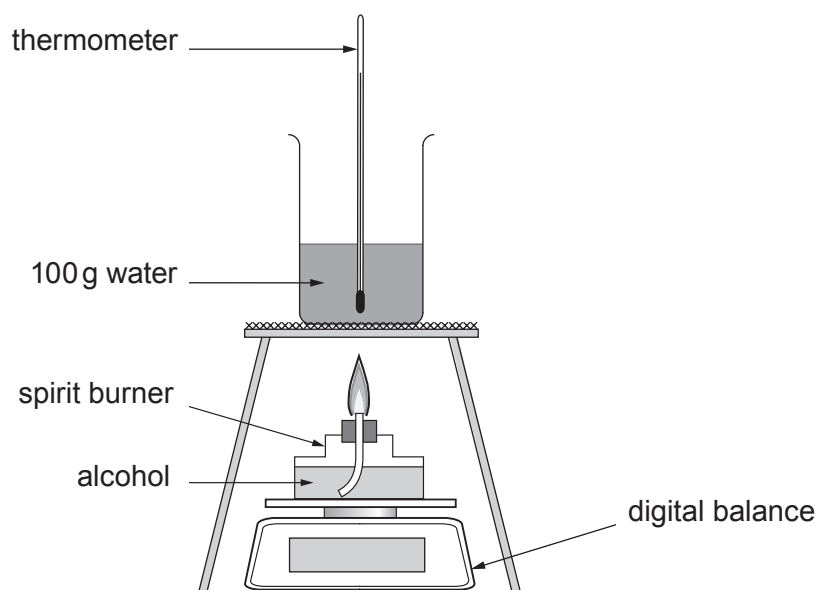
(ii) Name monomer **E**. [1]



E



- (c) A student carried out an investigation to find which alcohol gives out the most energy when burned.



100 g of water was heated by burning 1 g of each alcohol.

Alcohol	methanol	ethanol	propanol
Formula	CH ₃ OH	C ₂ H ₅ OH	C ₃ H ₇ OH
Relative molecular mass (M_r)	?	46	60
Temperature of water before heating (°C)	20	21	20
Temperature of water after heating (°C)	45	58	60
Energy given out (J)	?	15 700	17 200

- (i) Calculate the relative molecular mass (M_r) of methanol, CH₃OH.

[2]

$$A_r(\text{H}) = 1 \quad A_r(\text{C}) = 12 \quad A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$



- (ii) The energy given out can be calculated using the formula:

$$\text{energy given out} = \text{mass of water} \times 4.2 \times \text{temperature change}$$

Use the data given to calculate the energy given out when burning methanol. [2]

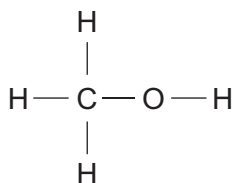
Energy given out = J

- (iii) Give the **letter** of the correct conclusion for the student's investigation. [1]

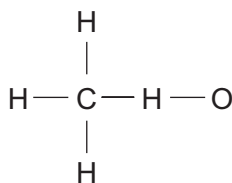
- A** all alcohols burn giving out the same amount of energy
- B** the greater the number of carbon atoms in the alcohol molecule the less energy is given out
- C** the greater the number of carbon atoms in the alcohol molecule the more energy is given out
- D** as the number of carbon atoms doubles the amount of energy given out doubles

Letter

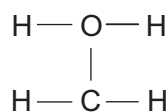
- (iv) Give the **letter** of the structural formula of methanol, CH₃OH. [1]



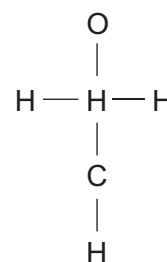
A



B



C

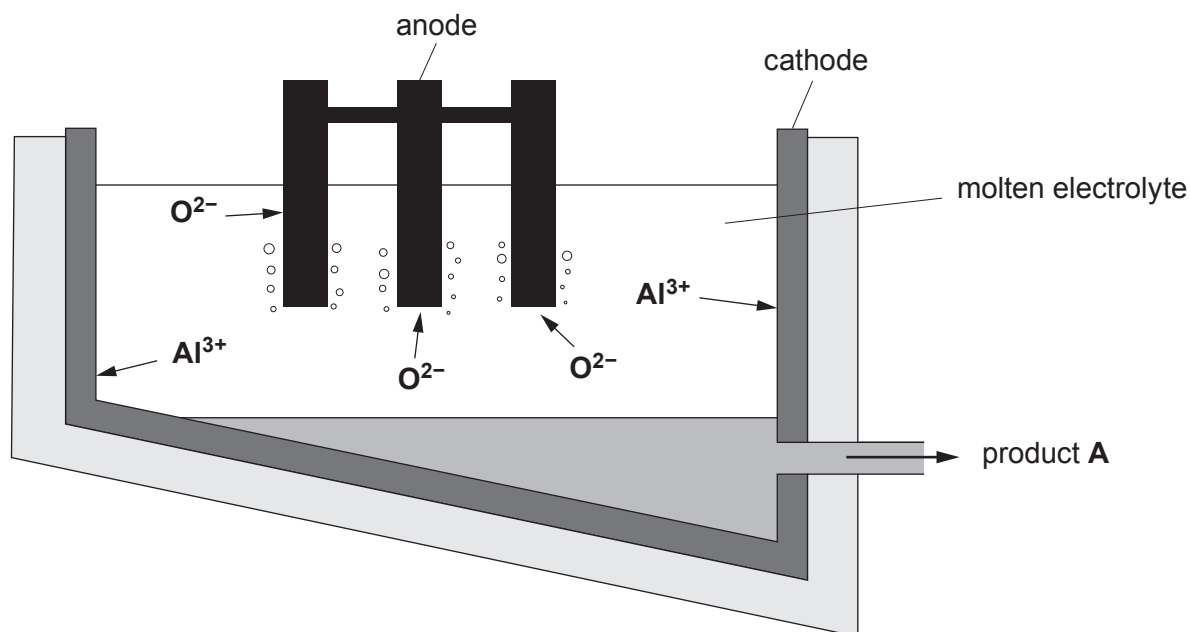


D

Letter



2. The diagram shows an electrolysis cell used in the extraction of aluminium from alumina.



bauxite	electrical	molten alumina	positive
negative	cryolite	molten aluminium	oxygen
			light

- (a) Choose words from the box to complete the following sentences. [5]

The molten electrolyte is a mixture of alumina and

Product **A** is

Oxygen is formed at the anode which is the electrode.

Alumina is obtained from an ore called

The process of extracting aluminium is expensive because it uses a lot of energy.

- (b) Balance the symbol equation that shows the overall reaction. [1]

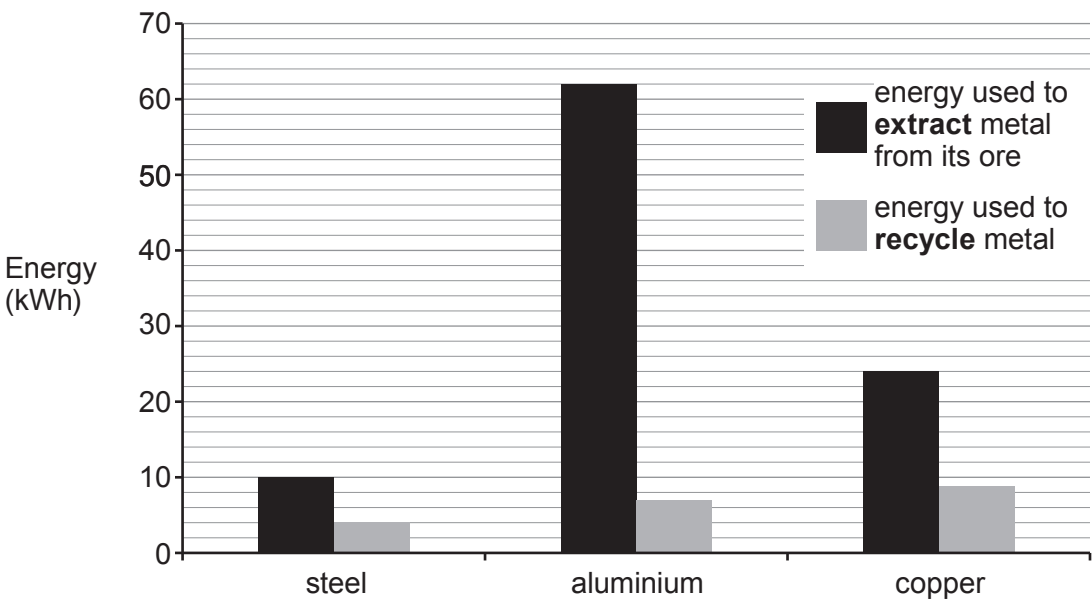


(c) 250 tonnes of an aluminium ore can produce 195 tonnes of alumina.
Calculate the percentage of alumina in the ore.

[2]

Percentage = %

(d) The chart compares the amount of energy used to extract and to recycle three metals.



Use the information in the chart to answer the following question.

Put a tick (✓) in the box next to the statement that describes why it is more cost effective to recycle aluminium than steel and copper. [1]

The energy used to extract metals is greater than that used in recycling them

☐

The difference between the energy used to extract and the energy used to recycle is the greatest

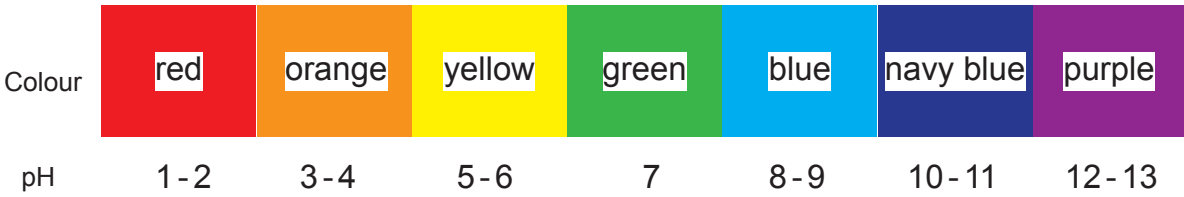
☐

The energy used in recycling is less than for copper but greater than for steel

☐

Examiner
only

2. (a) The diagram shows the colours of universal indicator at different pH values.



(i) Sodium hydroxide solution turns universal indicator purple.

Put a **tick** (✓) in the box next to the correct statement.

[1]

sodium hydroxide is a strong alkali

☐

sodium hydroxide is a weak alkali

☐

sodium hydroxide is a strong acid

☐

sodium hydroxide is a weak acid

☐

(ii) Ethanol is neutral. State the **colour** universal indicator would turn in ethanol.

[1]

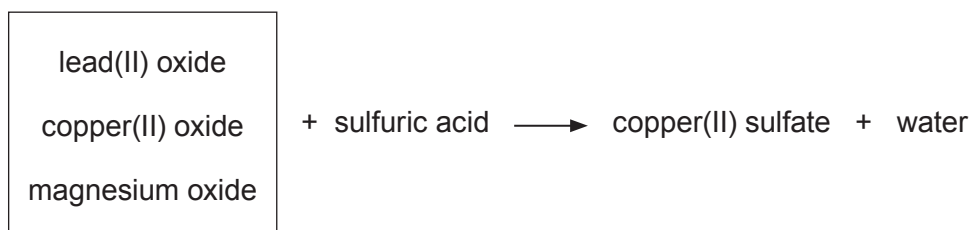
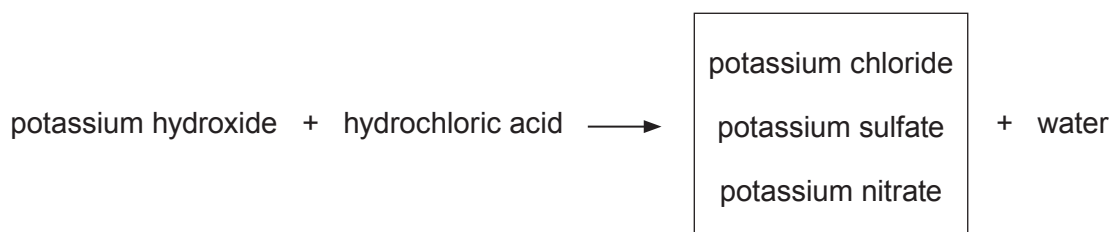
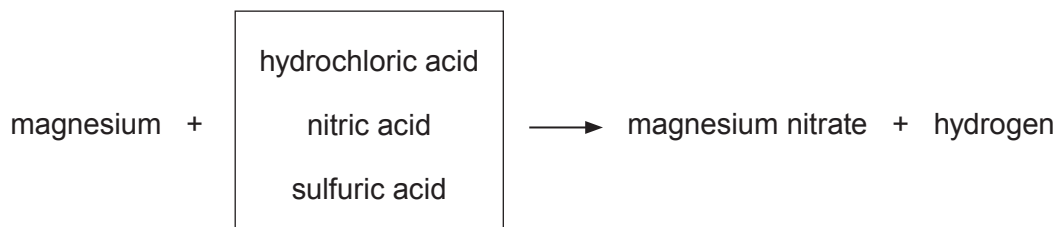
.....



(b) The word equations show reactions of acids with a metal, an alkali and a base.

Circle the correct substance in the box to complete each equation.

[3]



(c) A student was asked how he would prepare a sample of zinc chloride crystals from zinc.

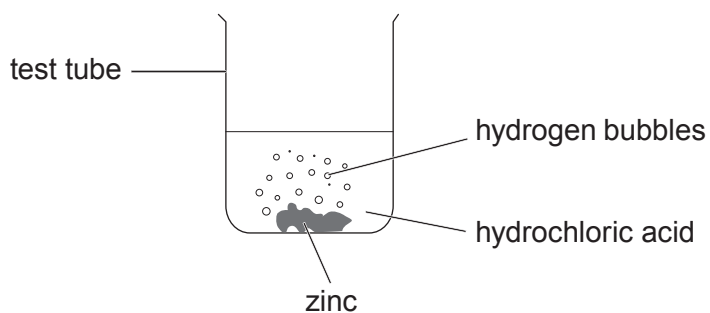
The word equation for the reaction is



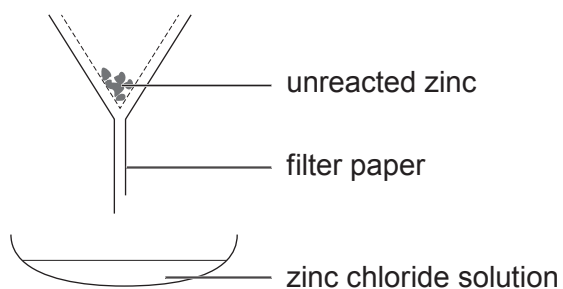
The diagrams below were taken from the student's work book.

There is **one** error in **each** of the student's diagrams.

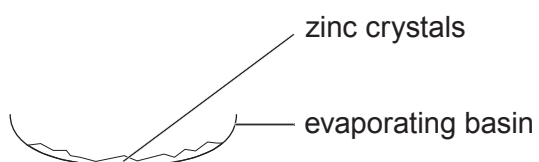
Step 1



Step 2



Step 3



(i) **Circle** the error in each diagram. [3]

(ii) Zinc chloride contains the ions Zn^{2+} and Cl^- .

Underline the correct formula of zinc chloride. [1]



(iii) Put a **tick** (✓) in the box next to the test to show that the gas given off is hydrogen. [1]

the gas pops with a lighted splint

☐

the gas turns limewater milky

☐

the gas relights a glowing splint

☐

2. (a) Crude oil is formed from simple marine organisms.

Tick (✓) the box next to the length of time it takes for the organisms to turn into crude oil.

[1]

hundreds of years

☐

thousands of years

☐

millions of years

☐

- (b) Crude oil can be separated into simpler mixtures called fractions.

Tick (✓) the box next to the method used to separate crude oil into fractions.

[1]

fractional distillation

☐

filtration

☐

cracking

☐

polymerisation

☐

(c) The table shows properties of some fractions that are obtained from crude oil.

Fraction	Size of molecules (chain length)	Viscosity	Ease of ignition	Amount of smoke formed
petrol	C ₅ –C ₁₀	very runny	very easy	no smoke
naphtha	C ₈ –C ₁₂	fairly runny	quite easy	little smoke
kerosene	C ₁₀ –C ₁₆	thick	quite hard	quite a lot of smoke
diesel oil	C ₁₄ –C ₂₀	very thick	very hard	very smoky

Use only the information from the table to answer the following questions.

- (i)

Name the fraction which is the easiest to pour.

[1]
- (ii)

Name the fraction which is the hardest to burn.

[1]
- (iii)

Name the fraction with the smallest **range** of chain lengths.

[1]

.....
- (iv)

Name the fraction which burns with the cleanest flame. Give the reason for your choice.

[2]

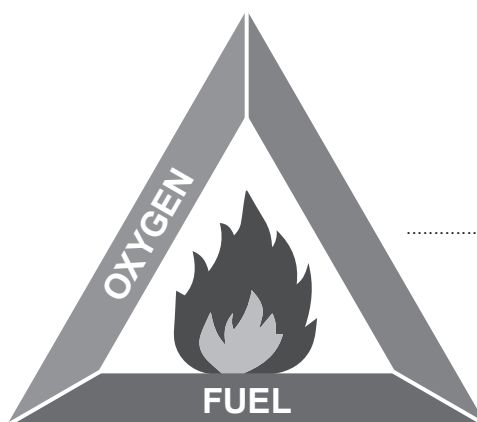
Fraction

Reason

3410U201
05



2. (a) In the diagram of the fire triangle, two sides have been labelled.



- (i) **Complete the fire triangle** by labelling the third side. [1]

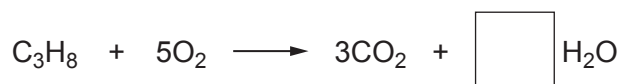
- (ii) Complete the following sentences to explain how a fire is extinguished by the following methods. [2]

Using a fire blanket removes the from the fire triangle.

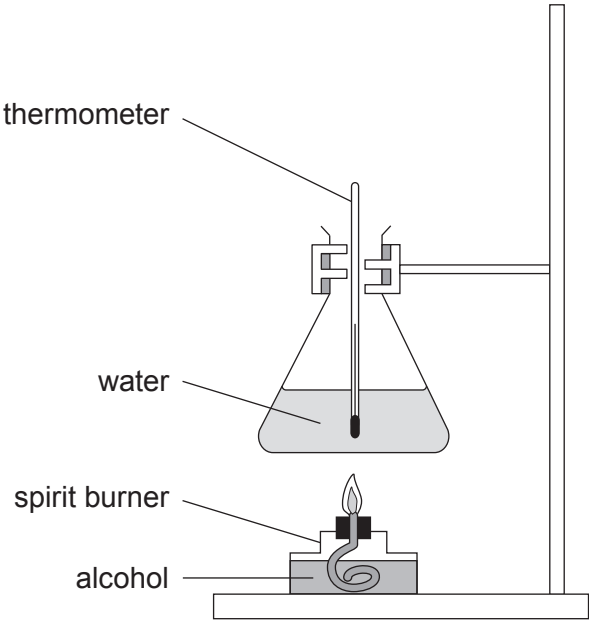
Cutting down trees in a forest fire removes the from the fire triangle.

- (b) Choose a **number** from the box below to balance the equation for the burning of propane gas. [1]

2	3	4	8
---	---	---	---



- (c) A class used the apparatus shown to compare the combustion of different alcohols. They burned each alcohol for 1 minute. They measured the temperature of the water before and after burning each alcohol.



The table shows the increase in temperature of the water for each alcohol.

Alcohol	Temperature increase (°C)
methanol	8
ethanol	19
propanol	23
butanol	38

- (i) The starting temperature of the water each time was 18 °C. Calculate the final temperature of the water after burning ethanol. [1]

Final temperature = °C



(ii) Tick (✓) the question that the class were trying to answer.

[1]

Which alcohol gives out the most heat energy?	
Which gases are produced when alcohols burn?	
Which alcohol has the lowest boiling point?	
Which alcohol burns for the longest?	

(d) Methanol has the chemical formula CH_3OH .

Calculate the relative molecular mass, M_r , of methanol.

[2]

$$A_r(\text{C}) = 12$$

$$A_r(\text{H}) = 1$$

$$A_r(\text{O}) = 16$$

$M_r =$

8

